Economic Order Quantity (EOQ)

order the right qty!

Cost and Management Accounting

In simple words *EOQ is the order size where ordering cost and carrying cost for the year are lowest together.*

1 EOQ

The order quantity that minimises the total of annual ordering cost and annual carrying cost of inventory.

At EOQ, ordering cost = carrying cost

2 Carrying cost

Cost of holding stock: storage, insurance, obsolescence, interest on funds blocked. Stated per unit per year.

Convert % to ₹ per unit per year

1. Total ordering cost for the year rises when orders are many (small order size)
2. Total carrying cost (Q/2 × C) rises when the order size is big
3. Cost per order (O) stays the same whatever the order size
4. At EOQ, total ordering cost equals total carrying cost



Swipe for the formula →

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THE EOQ FORMULA

Economic Order Quantity (EOQ)

one root, C at the bottom

Economic Order Quantity

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}}$$

A = annual demand (units); O = cost per order (₹); C = carrying cost per unit per year (₹)

Total annual ordering cost

$$ORDERING\ COST = (A / Q) \times O$$

Q = order quantity (units); A/Q = number of orders in the year

Total annual carrying cost

$$CARRYING\ COST = (Q / 2) \times C$$

Q/2 = average stock held, assuming uniform usage and no safety stock

MANY SMALL ORDERS OR FEW BIG ONES? EOQ balances the two: more orders cut carrying cost but raise ordering cost, and the reverse.

Swipe for a memory trick ⟶

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STUDY TRICK

Economic Order Quantity (EOQ)

picture it in words

2-A-O OVER C, ONE ROOT

Top of the fraction: 2, A, O multiplied. Carrying cost C is the divisor at the bottom. Put the whole fraction under one square root.

HOW TO ATTEMPT IT

1 Find A in units per year → 2 Find O per order →

3 Make C ₹ per unit per year → 4 Put 2AO / C under the root →

5 Check units of the answer

EXAM TIP

If carrying cost is given as a % of unit price, first multiply it by price to get C in ₹ per unit per year.

DOUBT BUSTER

Why is carrying cost Q/2 and not Q? Stock falls from Q to zero before the next lot arrives, so the average stock held is Q/2, assuming uniform usage.

Swipe for common mistakes →

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COMMON MISTAKES

Economic Order Quantity (EOQ)

avoid these slips

✗ Using monthly demand as A with a yearly C

✓ Keep A and C on the same period: annual demand and carrying cost per unit per year.

✗ Putting C in % of price directly in the formula

✓ Convert it first: C = % × price per unit, in ₹ per unit per year.

✗ Writing EOQ = √(2AC / O)

✓ EOQ = √(2AO / C): carrying cost is the divisor.

✗ Saying EOQ is where total cost is highest

✓ EOQ is where total of ordering and carrying cost is lowest.

TOMORROW'S CA INTER MCQ TESTS THIS! Revise it tonight, then try the MCQ.

Save this for revision!

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